

Vibration phenomena



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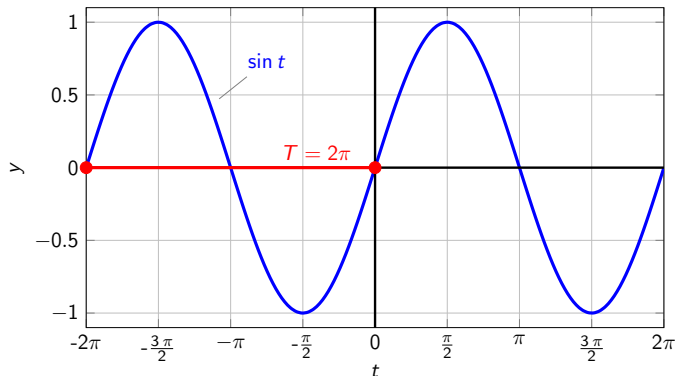


Introduction: Vibration phenomena

- ▶ In this section we use sine and cosine to describe phenomena that **oscillate in time**
- ▶ We introduce **elementary vibrations** and the quantities that characterize them: **period**, **amplitude**, **angular frequency**, **frequency** and **phase**
- ▶ We will see that the superposition of two vibrations with the same angular frequency is again an elementary vibration

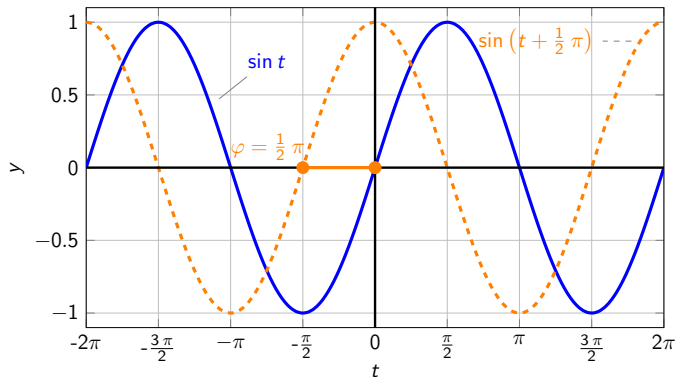
Sine and cosine in time

- We have seen that the sine and cosine functions are periodic with **period** 2π . We will use the variable t to denote time.



Phase shift

- Recall that $\sin(t + \varphi)$ corresponds to a **phase shift** φ (horizontal translation).



Elementary vibrations

The functions:

$$t \mapsto a \sin(\omega t), \quad t \mapsto b \cos(\omega t)$$

where a , b and ω are positive real numbers, are called **elementary vibrations**.

They are periodic functions with

$$\text{period} \quad T = \frac{2\pi}{\omega}$$

Example: period

For example, considering $t' = t + \frac{2\pi}{\omega}$:

$$\sin \left[\omega \underbrace{\left(t + \frac{2\pi}{\omega} \right)}_{t'} \right] = \sin(\omega t + 2\pi) = \sin \omega t$$

► Clearly we have:

$$|a \sin(\omega t)| \leq a \quad \text{and} \quad |b \cos(\omega t)| \leq b$$

Amplitude, angular frequency and frequency

The elementary vibrations $a \sin(\omega t)$ and $b \cos(\omega t)$ are characterized by

amplitude a and b (respectively)

$$\text{angular frequency } \omega = \frac{2\pi}{T}$$

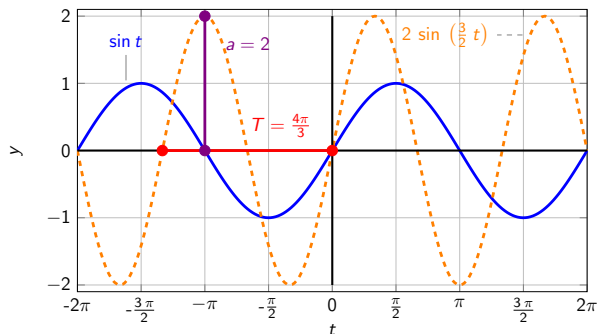
also called *angular velocity*, which indicates how many periods there are in an interval of length 2π . Moreover, they are characterized by the

$$\text{frequency } \nu = \frac{\omega}{2\pi}$$

which indicates how many times the function repeats itself in an interval of length 1.

Example: amplitude, angular frequency and frequency

$$t \mapsto 2 \sin \left(\frac{3}{2} t \right) \quad a = 2, \quad \omega = \frac{3}{2}, \quad T = \frac{4\pi}{3}$$



The angular frequency $\omega = \frac{3}{2}$ means that there are 1.5 periods in an interval of length 2π . The frequency ν is $\frac{3}{4\pi}$.

Superposition of vibrations (1/3)

- The function

$$h(t) = a \sin(\omega t) + b \cos(\omega t)$$

describes the **superposition** of the two elementary vibrations with the **same angular frequency** ω . The latter is again an elementary vibration, **phase-shifted** with respect to the previous ones.

- Indeed, setting $A = \sqrt{a^2 + b^2}$ we can write

$$h(t) = A \underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\alpha} \sin(\omega t) + A \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\beta} \cos(\omega t)$$

Superposition of vibrations (2/3)

- Now observe that the numbers

$$\alpha = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \beta = \frac{b}{\sqrt{a^2 + b^2}}$$

satisfy the conditions

$$-1 \leq \alpha \leq 1 \quad -1 \leq \beta \leq 1 \quad \alpha^2 + \beta^2 = 1$$

Hence there exists a **unique angle** φ such that

$$\cos \varphi = \alpha \quad \sin \varphi = \beta$$

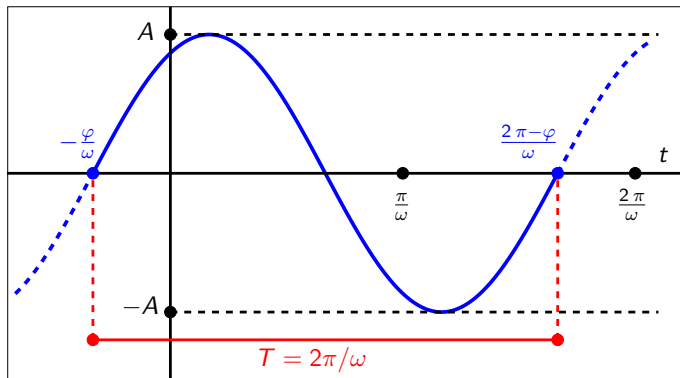
Superposition of vibrations (3/3)

- So that $h(t)$ can be rewritten in the following form

$$\begin{aligned}h(t) &= A \cos \varphi \sin(\omega t) + A \sin \varphi \cos(\omega t) \\&= A \sin(\omega t + \varphi)\end{aligned}$$

- Therefore $h(t)$ represents an elementary vibration with amplitude A , angular frequency ω , phase-shifted by an angle φ .

Superposition of vibrations: graph



In summary

$$a \sin(\omega t) + b \cos(\omega t) = A \sin(\omega t + \varphi)$$

$$A = \sqrt{a^2 + b^2} \quad \begin{cases} a = A \cos \varphi \\ b = A \sin \varphi \end{cases}$$

- Under fairly general conditions, a periodic natural phenomenon can be written as the superposition of a finite or infinite number of elementary vibrations with different frequencies (**Fourier series**).

Summary

Vibration phenomena:

- ▶ **Elementary vibrations:** $a \sin(\omega t)$ and $b \cos(\omega t)$, with period $T = \frac{2\pi}{\omega}$
- ▶ **Amplitude** a (or b), **angular frequency** $\omega = \frac{2\pi}{T}$, **frequency** $\nu = \frac{\omega}{2\pi}$
- ▶ $\sin(t + \varphi)$: **phase shift** φ
- ▶ $a \sin(\omega t) + b \cos(\omega t) = A \sin(\omega t + \varphi)$ with $A = \sqrt{a^2 + b^2}$
- ▶ General periodic phenomena: superposition of elementary vibrations (Fourier series)

Damping and amplification effects



Introduction: Damping and amplification effects

- ▶ By multiplying an elementary vibration by **powers** or **exponentials**, we can model **damping** or **amplification** effects.
- ▶ We see two examples: an **amplified vibration** $t \sin(\omega t)$ and a **damped vibration** $e^{-\alpha t} \sin(\omega t)$
- ▶ In both cases the graph lies between two curves that act as its “envelope”

First example (1/3)

- For example, consider the function

$$h(t) = t \sin(\omega t)$$

which models an **amplified vibration**.

- Since $-1 \leq \sin(\omega t) \leq 1$, we have

$$-t \leq t \sin(\omega t) \leq t$$

and hence the graph of $h(t)$ lies between the graphs of the lines with equations $y = -t$, $y = t$.

First example (2/3)

- At the points where $\sin(\omega t) = 1$, i.e.,

$$t = \frac{\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

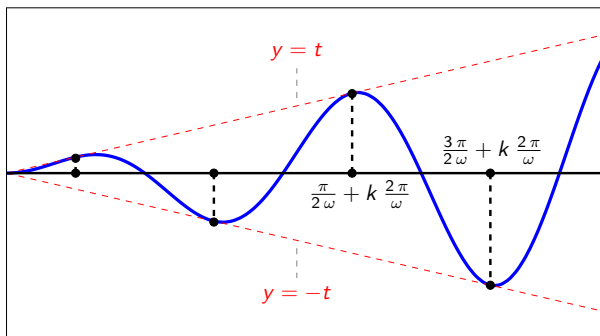
the graph of $h(t)$ touches that of $y = t$.

- At the points where $\sin(\omega t) = -1$, i.e.,

$$t = \frac{3\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

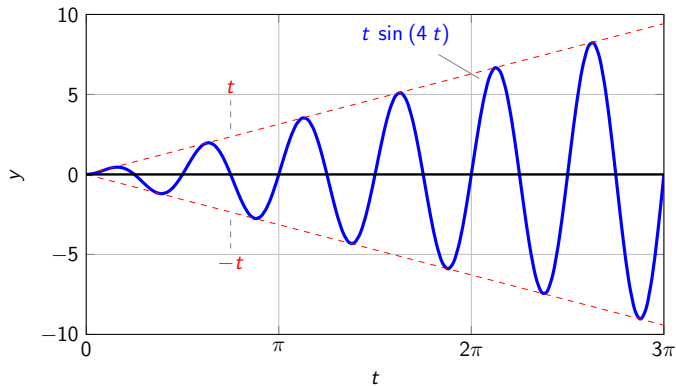
the graph of $h(t)$ touches that of $y = -t$.

First example (3/3)

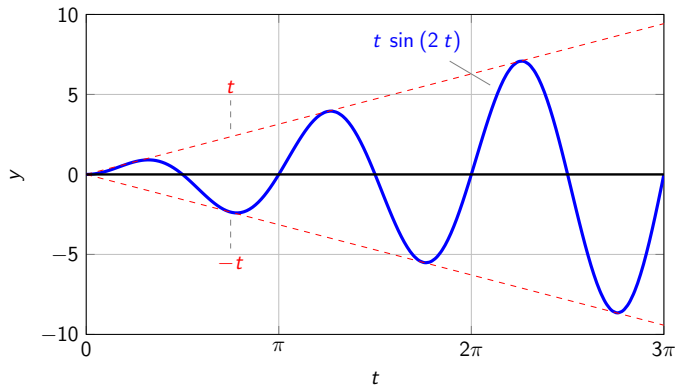


- From the graph we can see that multiplication by t has the effect of amplifying the vibration as t increases.

Amplified vibration: graphs (1/2)



Amplified vibration: graphs (2/2)



Second example

- ▶ For example, consider the function

$$k(t) = e^{-\alpha t} \sin(\omega t) \quad (\alpha > 0)$$

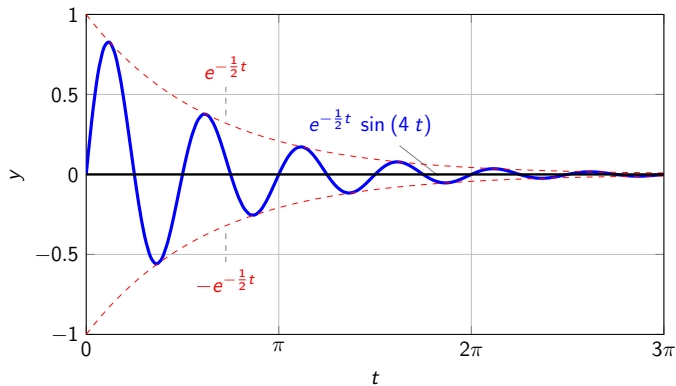
which models a **damped vibration**.

- ▶ Recalling that $e^{-\alpha t} = \left(\frac{1}{e^\alpha}\right)^t$ is an exponential with base less than 1, considerations analogous to those made for the function $h(t)$ show that the graph of k lies between the graphs of the functions

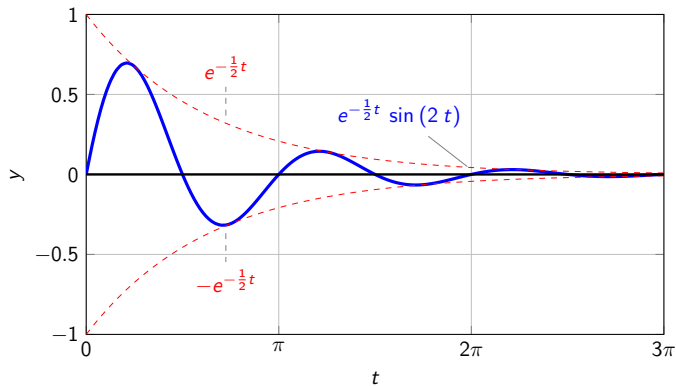
$$y_1 = e^{-\alpha t} \quad \text{and} \quad y_2 = -e^{-\alpha t}$$

as shown in the figure.

Damped vibration: graphs (1/2)



Damped vibration: graphs (2/2)



Summary

Damping and amplification:

- ▶ $h(t) = t \sin(\omega t)$: **amplified vibration**, with graph between $y = -t$ and $y = t$
- ▶ $k(t) = e^{-\alpha t} \sin(\omega t)$, $\alpha > 0$: **damped vibration**, with graph between $-e^{-\alpha t}$ and $e^{-\alpha t}$
- ▶ The graph touches the envelope curves at the points where $\sin(\omega t) = \pm 1$

Final summary

- ▶ **Elementary vibrations** $a \sin(\omega t)$, $b \cos(\omega t)$ are characterized by amplitude, angular frequency, period and frequency
- ▶ The superposition of two vibrations with the same angular frequency is a **phase-shifted** elementary vibration:
 $A \sin(\omega t + \varphi)$
- ▶ Multiplying by powers or exponentials gives **amplified** or **damped** vibrations